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COURSE NAME: CLOUD COMPUTING FOR BIG DATA ANALYTICS

ASSESSMENT TOOL-6

SCENARIO-BASED

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Ans) Cloud Architecture for Video Streaming

A suitable architecture for a video streaming platform should use **load balancing, auto-scaling, caching, and content delivery networks (CDNs)**.

Architecture:

Users → DNS → CDN → Load Balancer → Auto-Scaling Web/Streaming Servers → Cloud Storage + Database

- **CDN:** Stores frequently accessed video content at edge locations close to users. This reduces latency and buffering.
- **Load Balancer:** Distributes incoming requests among multiple servers so that no single server becomes overloaded.
- **Auto-Scaling:** Automatically increases the number of servers when the number of users increases and reduces them when demand decreases.
- **Cloud Storage:** Stores large video files using scalable object storage.
- **Caching:** Frequently watched videos are cached at CDN edge locations for faster delivery.
- **Database:** Stores user information, video metadata, subscriptions, and viewing history.

Load Balancing and Auto Scaling:

The load balancer continuously distributes traffic among available instances. During peak

hours, auto-scaling launches additional instances. When demand decreases, unnecessary instances are removed. This provides smooth streaming and high availability.

Cost optimization:

1. Use auto-scaling to avoid paying for unused resources.
2. Use CDN caching to reduce bandwidth and origin-server load.
3. Use suitable storage classes for frequently and rarely accessed videos.
4. Use reserved/committed resources for predictable workloads.
5. Monitor resource usage and remove unused resources.

Thus, scalability, caching, load balancing, and auto-scaling allow the platform to support millions of concurrent users with minimum buffering. scenario.docx

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Ans) Containerized Fintech Application Using Multi-Cloud

The fintech startup can use **Kubernetes-based container orchestration** across multiple cloud providers.

Proposed architecture:

Users → Global Load Balancer → Cloud Provider 1 / Cloud Provider 2 → Kubernetes Clusters → Containerized Applications

- Package applications into **Docker containers** to maintain consistency across cloud platforms.
- Use **Kubernetes** to deploy, manage, and scale containers.
- Use Kubernetes manifests/Helm charts to maintain consistent deployments.
- Use **service discovery** through Kubernetes Services and DNS.
- Use horizontal auto-scaling to increase or decrease containers according to workload.

Service discovery:

Kubernetes Services provide stable names and IP addresses for application components. Applications can locate services without depending on changing container IP addresses.

Scaling:

Kubernetes Horizontal Pod Autoscaler can increase or decrease application instances based on CPU, memory, or other metrics.

Benefits of multi-cloud:

1. Avoids dependence on a single cloud provider.
2. Improves availability.
3. Provides flexibility to select services from different providers.
4. Can improve disaster recovery.
5. Provides greater geographic coverage.

Risks:

1. More complex management.
2. Different cloud APIs and services.
3. Higher operational costs.
4. Difficult monitoring and troubleshooting.
5. Data synchronization and security challenges.

Therefore, Kubernetes with standardized container images and deployment configurations provides consistency and efficient multi-cloud management. scenario.docx

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Ans) Backup and Disaster Recovery for E-Learning Platform

The organization should implement a **cloud-native backup and disaster recovery system** using backup, replication, and versioning.

Proposed solution:

Application → Primary Cloud Storage/Database → Automated Backup → Replicated Secondary Region

- Enable **versioning** so that previous versions of deleted or modified files can be restored.
- Schedule automatic backups at regular intervals.
- Replicate important data to a different availability zone or region.
- Keep multiple backup copies to protect against accidental deletion and failures.
- Encrypt backup data both during transmission and while stored.

RTO and RPO

RTO (Recovery Time Objective):

Defines the maximum acceptable time required to restore the service after a failure.

RPO (Recovery Point Objective):

Defines the maximum acceptable amount of data loss measured in time.

For example, if the RPO is 15 minutes, backups/replication should ensure that no more than approximately 15 minutes of data is lost.

Business continuity steps:

1. Detect the failure.
2. Activate the disaster recovery system.
3. Redirect users to the backup/secondary environment.
4. Restore the latest valid backup.
5. Verify application and database integrity.
6. Continue services from the recovery environment.
7. Restore the primary environment after fixing the failure.
8. Regularly test the disaster recovery plan.

This approach minimizes data loss and downtime while maintaining continuous e-learning services. scenario.docx

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Ans) Hybrid Cloud for Government Agency

The government agency should use a **hybrid cloud architecture** where sensitive workloads remain on-premises while suitable non-sensitive workloads operate in the public cloud.

Architecture:

Users → Secure Gateway → Hybrid Network

On-Premises:

Sensitive data + Compliance-critical applications

Public Cloud:

Non-sensitive applications + Scalable services

The two environments should be connected using a **secure VPN or dedicated private connection**.

Identity Management

- Use centralized identity and access management.

- Implement **role-based access control (RBAC)**.
- Apply least-privilege access.
- Use multi-factor authentication for privileged users.
- Maintain centralized authentication and authorization policies.

Encryption

- Encrypt sensitive data while stored using encryption-at-rest.
- Use TLS encryption for data transmitted between on-premises and cloud environments.
- Secure and regularly rotate encryption keys.
- Protect backups using encryption.

Challenges

1. Managing two different environments.
2. Maintaining consistent security policies.
3. Network complexity.
4. Identity and access management across environments.
5. Monitoring and troubleshooting.
6. Compliance and regulatory requirements.
7. Data synchronization between environments.

Therefore, a secure hybrid architecture allows the agency to retain sensitive workloads on-premises while using the scalability and flexibility of the cloud. scenario.docx

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Ans) Multi-Region SaaS Deployment

The SaaS provider should deploy its application across **multiple geographic cloud regions** to provide high availability and fault tolerance.

Architecture:

Users → Global DNS/Traffic Manager → Region 1 / Region 2 → Application Servers → Databases

- Deploy application servers in at least two separate regions.

- Use a global load balancer or traffic manager to route users to healthy regions.
- Configure automatic health checks.
- Replicate databases between regions.
- Maintain backups in independent locations.

Data Replication

Data should be continuously or periodically replicated from the primary region to the secondary region.

For critical applications, **synchronous replication** can reduce data loss where supported, while **asynchronous replication** can provide better geographic flexibility and performance.

Failover

1. Continuously monitor the health of each region.
2. Detect failure using health checks.
3. Automatically redirect traffic to the healthy region.
4. Promote the secondary database when required.
5. Restore the failed region after the problem is resolved.
6. Synchronize data before returning it to normal operation.

Monitoring and Alerting

Monitor:

- CPU and memory usage.
- Application response time.
- Error rates.
- Database health.
- Network performance.
- Availability of each region.
- Replication lag.

Configure alerts for abnormal conditions such as increased errors, high latency, resource exhaustion, or regional failure. Automated monitoring enables administrators to identify problems early and take corrective action.

Thus, multi-region deployment with replication, automated failover, and continuous monitoring ensures high availability even when one region becomes unavailable.